Patrick Thangarajah

DATA ANALYSIS | STATISTICAL MODELLING | SIMULATION | TECHNICAL COMMUNICATION

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Summary

Completed an MASc in Mechanical and Mechatronics Engineering at the University of Waterloo in 2026, achieving a 90% graduate coursework average. Developed analytical, statistical modelling and communication skills through simulation studies, physical test validation, machine learning and sponsor presentations, including to Transport Canada, across medical devices, vehicle safety and manufacturing.

Skills

Data Analysis & Statistics	Python, MATLAB, JMP, DOE, least-squares regression, ANOVA, Microsoft Excel
Model Validation and Testing	FE model validation, physical test correlation, ISO 18571, experimental data analysis
Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Research in impact mechanics, finite element analysis, and model validation

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*

- Designed and analyzed a 104-simulation study using JMP DOE, least-squares regression and ANOVA, identifying a seat belt anchor position that reduced peak chest deflection by 16% relative to the study average
- Validated a THOR-5F finite element model against physical sled tests in 2 configurations, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Investigated discrepancies between simulation predictions and experimental results by evaluating material response, contact, boundary conditions, occupant positioning and restraint fit
- Developed LS-PrePost scripts that streamlined a repetitive simulation setup from 7 steps to 4, reducing setup steps by 43%
- Presented research findings through a master's thesis, conference paper, technical reports and six sponsor presentations, including to Transport Canada

Summer Research Student — *Vancouver, BC, Canada (Remote)*
Centre for Heart Lung Innovation — *May 2022 - Aug 2022*

- Built a transfer-learning diagnostic model in a Jupyter Notebook, improving diabetic kidney disease detection accuracy by 10% through optimization, training refinement and tissue-slide image comparisons

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*

- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*

- Reduced fiberglass cutting waste by 50% through process analysis and proposed a material handling workflow projected to halve pick-and-place time
- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes

Projects

Simulation of Cervical Fractures in American Football (ME 720 Course Project) — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.